

Guanlin Yu

US Citizen, no need for VISA sponsorship.

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Seattle WA(Open to Relocation)

Portfolio: <https://www.guanlinyu.com>

Work Experience

Reality AI • Software Developer Intern01/2025 – present(Remote)

- Partnered with cross-functional teams to design and implement AI-driven solutions aligned with business goals, leveraging Java and **TensorFlow**.
- Performed in-depth data analysis and preprocessing to deliver high-quality inputs for machine learning models, utilizing **Pandas** for data manipulation.
- Designed, implemented, and fine-tuned machine learning algorithms, achieving over a 25% improvement in accuracy using frameworks such as **PyTorch**.
- Deployed end-to-end AI systems, seamlessly integrating them with existing infrastructure, ensuring scalability through **Docker** containerization.

OSCR AI • Software Developer InternLA, California, USA10/2024 – 2/2025

Marketing Integration Project

- Built and integrated a **React**-based frontend component to enable affiliate marketing conversion tracking, improving business analytics capabilities.
- Enhanced marketing performance visibility by enabling tracking metrics for conversion rates, leading to a 20% improvement in affiliate sales conversion efficiency.
- Used React hooks for **state management** to ensure seamless data flow and React Router for navigation within the app.
- Conducted **API integration testing** using **Postman** and debugged issues using Chrome DevTools, reducing errors by 15%.
- Collaborated with cross-functional teams to align affiliate integration with business goals, completing the project 10% ahead of schedule.

Full-Stack AI-Powered Q&A Platform for PDF Document Queries10 – 12/2024

- Developed a full-stack React-based interactive web application using Ant Design, enabling seamless uploading and querying of PDF documents.
- Designed and implemented a robust backend with **Express.js** and **Multer** for file uploads and efficient API handling.
- Leveraged **LangChain** and **OpenAI GPT APIs** to load, split, and embed PDF content into a **memory vector** store for **real-time document-based query retrieval**.
- Featured both real-time voice interaction and speech/script interaction between user and web app.
- Integrated a Retrieval QA Chain for advanced **AI-driven responses**, ensuring high accuracy and scalability.
- Optimized the system for efficient processing of large documents and **natural language queries**.

SCEGC ENGINEERING DESIGN INSTITUTE • Web App Developer Intern

Employee Timekeeping App Based on Facial Recognition Project12/2023-02/2024(Remote)

- Utilized Streamlit to design and implement an easy-to-use user front-end interaction page.
- Core function utilized **MTCNN**, **InceptionResnetV1** model from facenet **Pytorch Machine Learning** library to obtain and compare the facial feature data.
- Solved the write-heavy backend need by integrating **PostgreSQL** driver psycopg2 to connect the database and update local and database data in both directions—improved timekeeping accuracy, decreasing timekeeping error follow-up rate by 30%.

SELECTED PROJECTS

Audio Media Android Project09/2024-11/2024

- Designed a **Kotlin** Spotify-like favorite app by fully leveraging **Android Jetpack Library** and **Hilt Dependency Injection**.
- Implemented BottomBar and App Navigation using **Jetpack Navigation** components.
- Created Restful Api json-server and used **Retrofit** to handle requests.
- Built the feed/album/favorite UI components in **Jetpack Compose** following **MVVM** structure.
- Enabled local cache ability for favorite feature feed by using **Room** database.
- Integrated the **Google Exoplayer** to handle global music playback.

E-commerce Flash Sell Project03/2024-07/2024

- Implemented a **distributed ID system (Snowflake ID algorithm)** to ensure unique order identifiers. Ensured unique order numbers while also enabling the query based on time stamps.
- Leveraged **Redis** for cache warming to reduce database load and improve response time.
- Ensured data consistency between payment and inventory reduction operations by first processing order inventory count in the cache. Incorporated **SpringBoot Segmented Locking** feature, separating order inventory deduction in different phases of ordering, improving system response during high traffic spikes.
- Utilized message middleware to smooth traffic spikes, integrating order processing with the **Message queue (Rocket MQ)** autoscaling feature. Stabilized database query traffic by Peak Loading Shifting using Lua script-enabled **asynchronous tasks**.
- Employed **Content Delivery Network (CDN)** for traffic distribution and static page rendering to alleviate server load.Bypassed dynamic page rendering, thus increasing performance by 60% under traffic spike.
- Integrated **Sentinel traffic control framework** with Spring Boot to manage and control traffic flows effectively.
- Utilized Jmeter to test traffic flow, ensuring system performance at 10,000 queries/second level.

Skills

Programming Language

Java | Python | Go |
JavaScrip | C# | C++

Tools & Frameworks:

Spring/SpringBoot | Node.js |
Django | Kubernetes | gRPC |
Protocol Buffer | React | HTML |
CSS | SQL(MySQL) |
NoSQL(Redis, Memcached) |
Message Queue(AWS SQS,
RocketMQ, Kafka) | Git | Agile

AWS

S3 | SNS | SQS | EC2 | ECS |
DynamoDB | CloudWatch |
API Gateway | Lambda function

Platform/OS

Linux | MacOS | Windows

Languange

English	Native
Chinese	Native
French	Fluent
Japanese	Fluent

EDUCATION

M.S.
Computer Information System
Bellevue University
Cumulative GPA: 3.92
12/2025(Expected)

B.S.
Pre Med
University of Washington
(Seattle, WA)
05/2021